# claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

## Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-aefc5dfe02f97eaee.jsonl`
- SHA-256: `489eea01948eb74ac9d0e22faa9518d42840f3c19eac2f6397ec9f212fea04a1`
- Source modified: `2026-06-06T17:37:24+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

## Transcript

#### 1. user (2026-06-06T17:35:30.297Z)

## Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

## Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

### Claim under review

"Self-supervised learning combined with knowledge distillation is a proven, top-tier approach for action spotting: COMEDIAN (SSL + KD) achieved 73.10% tight mAP on SoccerNet, validating teacher->student distillation as a strong direction rather than a workaround."

**Source:** https://arxiv.org/html/2505.03991v1 (primary)
**Supporting quote:** "First AS-specific framework that combines self-supervised learning (SSL) with knowledge distillation (KD)"

### Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

**refuted=true** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
**refuted=false** ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

#### 2. assistant (2026-06-06T17:35:34.454Z)

I'll verify this claim about COMEDIAN (SSL + KD) achieving 73.10% tight mAP on SoccerNet. Let me start by examining the primary source and searching for corroborating/contradicting evidence.

#### 3. assistant (2026-06-06T17:35:37.801Z)

Now let me research this claim. I'll fetch the primary source and search for the COMEDIAN paper's actual numbers.

#### 4. user (2026-06-06T17:35:45.328Z)

## COMEDIAN in this Survey

Yes, this paper mentions COMEDIAN. According to the document:

**COMEDIAN's Approach:**
The paper states that COMEDIAN is "the first AS-specific framework that combines self-supervised learning (SSL) with knowledge distillation (KD) for initializing spatiotemporal transformers." It features two Transformers: one for short-range context and another for longer-range context, with training using Momentum Contrast and Soft Contrastive loss.

**Performance Results:**
On SoccerNet-v2, COMEDIAN achieved:
- **Tight mAP: 73.10** (Test Set)
- **Tight mAP: 68.38** (Challenge Set)
- **Loose mAP: 73.98** (Challenge Set)

**About This Survey:**
This paper is a comprehensive review of deep learning methods for video event detection in sports, covering three tasks: Temporal Action Localization (TAL), Action Spotting (AS), and Precise Event Spotting (PES). Rather than presenting original results, the survey analyzes

existing datasets, methods, evaluation metrics, and identifies challenges like poor cross-sport generalization and reliance on extensive labeled data.

**5. user (2026-06-06T17:35:48.116Z)**

Web search results for query: "COMEDIAN self-supervised knowledge distillation action spotting SoccerNet tight mAP 73.10"

Links: [{"title":"COMEDIAN: Self-Supervised Learning and Knowledge Distillation for Action Spotting Using Transformers | IEEE Conference Publication | IEEE Xplore","url":"https://ieeexplore.ieee.org/document/10495649/"},{"title":"[2309.01270] COMEDIAN: Self-Supervised Learning and Knowledge Distillation for Action Spotting using Transformers","url":"https://arxiv.org/abs/2309.01270"},{"title":"Papers with Code - COMEDIAN: Self-Supervised Learning and Knowledge Distillation for Action Spotting using Transformers","url":"https://paperswithcode.com/paper/comedian-self-supervised-learning-and"},{"title":"[2309.01270] COMEDIAN: Self-Supervised Learning and Knowledge Distillation for Action Spotting using Transformers","url":"https://ar5iv.labs.arxiv.org/html/2309.01270"},{"title":"COMEDIAN: Self-supervised learning and knowledge distillation for action spotting using transformers - Direction de la recherche technologique","url":"https://cea.hal.science/DRT/hal-04462159v1"},{"title":"COMEDIAN: Self-supervised learning and knowledge distillation for action spotting using transformers - Archive ouverte HAL","url":"https://hal.science/hal-04462159"},{"title":"COMEDIAN: Self-supervised learning and knowledge distillation for action spotting using transformers - GREYC image","url":"https://memsic.ccsd.cnrs.fr/GREYC-IMAGE/hal-04462159v1"},{"title":"COMEDIAN: Self-Supervised Learning and Knowledge Distillation for Action","url":"https://arxiv.org/pdf/2309.01270"},{"title":"SoccerNet 2023 Challenges Results","url":"https://arxiv.org/pdf/2309.06006"},{"title":"COMEDIAN: Self-Supervised Learning and Knowledge Distillation for Action Spotting using Transformers - Julien Denize","url":"https://juliendenize.github.io/publication/comedian"}]

The search results provide comprehensive information about COMEDIAN. Here's what I found:

## COMEDIAN Overview

COMEDIAN is a novel pipeline to initialize spatiotemporal transformers for action spotting, which involves self-supervised learning and knowledge distillation. Action spotting is a timestamp-level temporal action detection task.

## Pipeline Architecture

The pipeline consists of three steps, with two initialization stages. First, self-supervised initialization of a spatial transformer using short videos as input. Additionally, a temporal transformer is initialized that enhances the spatial transformer's outputs with global context through knowledge distillation from a pre-computed feature bank aligned with each short video segment. In the final step, the transformers are fine-tuned to the action spotting task.

## Performance

The experiments, conducted on the SoccerNet-v2 dataset, demonstrate state-of-the-art performance and validate the effectiveness of COMEDIAN's pretraining paradigm. COMEDIAN achieved SOTA (State-of-the-Art) for Action Spotting on SoccerNet-v2 using the Tight Average-mAP metric.

## Publication

The work was published as "COMEDIAN: Self-supervised learning and knowledge distillation for action spotting using transformers" at the WACV 2024 - IEEE/CVF Winter Conference on Applications of Computer Vision, in January 2024, Waikoloa, United States.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

**6. assistant (2026-06-06T17:35:51.905Z)**

The claim's numbers are confirmed by the survey. Now let me verify against the primary COMEDIAN paper itself, and critically check the framing ? whether "73.10% tight mAP" is genuinely "top-tier" and whether SSL+KD is what drove that result, or whether the claim overreaches in extrapolating to general "validation of teacher->student distillation."

**7. user (2026-06-06T17:35:57.732Z)**

I cannot provide the specific numbers you're requesting based on the content provided. The webpage shows only the abstract and metadata for the COMEDIAN paper (arXiv:2309.01270), which states:

"Our pipeline consists of three steps, with two initialization stages" and demonstrates "state-of-the-art performance," but does not include:

- Specific tight Average-mAP scores on the test set
- Prior state-of-the-art baseline numbers or improvement margins
- Detailed input modalities or feature specifications
- Model size or computational requirements

To obtain these quantitative results, performance comparisons, and technical specifications, you would need to access the full PDF of the paper itself, which is linked on the page but not reproduced in the content provided.

**8. user (2026-06-06T17:36:06.620Z)**

Web search results for query: "SoccerNet action spotting tight mAP state of the art leaderboard 2024 2025 best methods T-DEED"

Links: [{"title":"SoccerNet 2024 Challenges Results","url":"https://arxiv.org/pdf/2409.10587"},{"title":"SoccerNet Dataset: Sports Video Benchmark","url":"https://www.emergentmind.com/topics/soccernet-dataset"},{"title":"SoccerNet - Action Spotting","url":"https://www.soccer-net.org/tasks/action-spotting"},{"title":"SoccerNet 2025 Challenges","url":"https://www.emergentmind.com/topics/soccernet-2025-challenges"},{"title":"SoccerNet 2025 Challenges Results","url":"https://arxiv.org/pdf/2508.19182"},{"title":"SoccerNet 2025 Challenges Results","url":"https://arxiv.org/html/2508.19182v1"},{"title":"SoccerNet - Ball Action Spotting","url":"https://www.soccer-net.org/tasks/ball-action-spotting"},{"title":"Action Spotting using Dense Detection Anchors Revisited: Submission to  the SoccerNet Challenge 2022","url":"https://arxiv.org/pdf/2206.07846"}]

Based on the search results, here's what I found about SoccerNet action spotting state-of-the-art performance:

## 2024 Challenge Results

In the 2024 ball action spotting challenge, 8 teams participated with a total of 61 submissions, with an improvement of mAP@1 from 56.15 for the baseline to 73.39 mAP@1 for the winning method.

The winning method was T-DEED, which achieved 73.39 mAP@1 and 77.25 tight avg-mAP. The second-place team (UniBw Munich - VIS) achieved 71.35 mAP@1 and 75.24 tight avg-mAP.

## 2025 Challenge Results

The SoccerNet 2025 Challenges comprise the fifth annual instantiation, with four major vision-based tasks including Team Ball Action Spotting, Monocular Depth Estimation, Multi-View Foul Recognition, and Game State Reconstruction.

For the 2025 challenges, most top entries began with the T-Deed baseline for ball action spotting, with innovations primarily centered on fusion of team information. The winning approach abandoned separate heads for action class and team assignment in favor of a joint head that predicts team?action combinations directly.

### Key Metrics

Starting in 2022, the tight average-mAP was introduced as an evaluation metric, which uses a smaller temporal tolerance ([1-5] seconds instead of [5-60 seconds]).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

#### 9. assistant (2026-06-06T17:36:10.525Z)

Let me get the precise numbers from the COMEDIAN full text and the actual SoccerNet action-spotting leaderboard to assess whether 73.10 is genuinely "top-tier" and whether the gains are attributable to SSL+KD specifically.

#### 10. user (2026-06-06T17:36:19.178Z)

# COMEDIAN Performance Summary

## Test Set Results
According to Table 9, COMEDIAN achieved **72.0% t-AmAP** (ViViT-T ensemble) on SoccerNet-v2's test split. The previous best method, Spivak (2022), scored **65.1%**, representing a **+6.9 percentage point improvement**.

## Ablation Study Findings
Table 5 demonstrates the SSL+KD pretraining advantage:

- **No pretraining (scratch):** 48.1% t-AmAP
- **SoccerNet SSL only (Step 1):** 54.7%
- **SSL + KD (Steps 1+2):** 64.7%
- **ImageNet supervised (Step 1):** 57.7%

The data shows that combining SSL with KD substantially outperforms random initialization and nearly closes the gap with supervised ImageNet pretraining, validating the combined approach's effectiveness.

## End-to-End vs Feature-Based Methods
COMEDIAN is fully end-to-end, processing raw images rather than pre-extracted features. Notably, it decisively beats E2E-Spot (61.8%), the previous best end-to-end competitor, by **+10.2 percentage points**.

#### 11. user (2026-06-06T17:36:19.398Z)

Web search results for query: "SoccerNet-v2 action spotting tight average-mAP COMEDIAN 73.1 comparison SOTA ranking paperswithcode"

Links: [{"title":"Papers with Code - Action Spotting","url":"https://paperswithcode.com/task/action-spotting"},{"title":"SoccerNet-v2","url":"https://silviogiancola.github.io/SoccerNetv2/"},{"title":"Soccernet V2","url":"https://www.catalyzex.com/s/Soccernet%20V2"},{"title":"Camera Calibration and Player Localization in SoccerNet-v2 and   Investigation of their Representations for Action Spotting","url":"https://arxiv.org/pdf/2104.09333"},{"title":"Daily Papers - Hugging Face","url":"https://huggingface.co/papers?q=ball+action+spotting"},{"title":"SoccerNet-v2: A Dataset and Benchmarks for Holistic Understanding of   Broadcast Soccer Videos","url":"https://arxiv.org/pdf/2011.13367"},{"title":"GitHub - SoccerNet/sn-spotting: Repository containing all necessary codes to get started on the SoccerNet Action Spotting challenge. This repository also contains several benchmark methods. · GitHub","url":"https://github.com/SoccerNet/sn-spotting"},{"title":"SoccerNet - Action Spotting","url":"https://www.soccer-net.org/tasks/action-spotting"}]

Based on the search results, I found relevant information about COMEDIAN and SoccerNet-v2 action spotting:

COMEDIAN achieved the 1st place in the SoccerNet-v2 Action Spotting challenge with a substantial performance improvement of 1.6 points in avg-mAP compared to the runner-up methods. COMEDIAN is a novel pipeline to initialize spatio-temporal transformers for action spotting, which involves self-supervised learning and knowledge distillation.

While the search results confirm COMEDIAN's state-of-the-art performance on SoccerNet-v2 action spotting, the specific 73.1 tight average-mAP score and its detailed ranking comparison on Papers with Code were not explicitly mentioned in these results. However, the information confirms that COMEDIAN achieved top performance on this benchmark task with significant improvements over competing methods.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 12. assistant (2026-06-06T17:36:37.464Z)

I now have enough evidence to evaluate the claim rigorously. Let me assess each part.

The claim has three assertions:
1. **Factual accuracy of the benchmark number**: "COMEDIAN (SSL + KD) achieved 73.10% tight mAP on SoccerNet" ? CONFIRMED. The survey paper (the cited source 2505.03991) reports COMEDIAN at 73.10 tight mAP on SoccerNet-v2 test set. The primary paper's ablations corroborate the SSL+KD pipeline (ensemble reaches ~72-73), and it was 1st place / SOTA at the time (WACV 2024).

2. **"SSL + KD" is genuinely the mechanism**: SUPPORTED. The COMEDIAN ablation (Table 5) shows SSL?54.7, SSL+KD?64.7, clearly demonstrating KD adds value on top of SSL. The supporting quote ("First AS-specific framework that combines SSL with KD") is accurate.

3. **The framing ? "validating teacher->student distillation as a strong direction rather than a workaround" / "proven, top-tier approach"**: This is where the claim overreaches and is partially misleading for OUR context.

Let me verify the most important refutation angle: whether COMEDIAN is actually still "top-tier," and critically, whether its KD is even the same thing as the research question's "cloud-VLM teacher ? local-model distillation."

### 13. user (2026-06-06T17:36:48.451Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: https://paperswithcode.com/task/action-spotting
Redirect URL: https://huggingface.co/papers/trending
Status: 302 Found

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:
- url: "https://huggingface.co/papers/trending"
- prompt: "List the top methods on SoccerNet-v2 action spotting by tight average-mAP. Where does COMEDIAN rank? Are there methods that beat 73.10 tight mAP (e.g. T-DEED, UglyDuckling, or 2024/2025 challenge winners)? Give specific numbers and method names if available."

### 14. assistant (2026-06-06T17:37:04.923Z)

I already have strong evidence on the ranking. From the earlier search, T-DEED achieved **77.25 tight avg-mAP** in the 2024 challenge ? beating COMEDIAN's 73.10. So COMEDIAN was SOTA at WACV 2024 but has since been surpassed. This nuances "top-tier" but doesn't refute it (73.10 is still a strong, near-frontier number).

Let me consolidate. The key question for refutation is whether the claim is *fair and well-supported*, not whether COMEDIAN exists. The critical issue is the inferential leap in the claim's framing relative to the research context.

Key findings:

- **Number is accurate**: 73.10 tight mAP confirmed by the cited source (survey 2505.03991) and consistent with the primary paper (arXiv 2309.01270, WACV 2024).
- **SSL+KD is real and ablation-validated**: SSL 54.7 ? SSL+KD 64.7; supporting quote is accurate.
- **"top-tier"**: Was SOTA at WACV 2024 (1st place SoccerNet-v2, +1.6 over runner-up). By the 2024 challenge, T-DEED hit 77.25 tight avg-mAP, surpassing it. So "top-tier" is defensible for its era but no longer the frontier ? a mild overstatement of currency but not a fabrication.
- **The load-bearing overreach**: COMEDIAN's "knowledge distillation" is NOT "teacher?student distillation" in the sense the research question uses it (a cloud VLM teacher labeling rallies ? a small local student). COMEDIAN's KD is *self-distillation from a pre-computed feature bank* to initialize a temporal transformer during self-supervised pretraining ? an in-domain, same-modality representation-learning trick, not a large-teacher-labels-then-small-student-learns-the-task paradigm. The claim uses COMEDIAN to "validate teacher?student distillation as a strong direction" for the rally-segmentation pipeline, but the two "distillations" are categorically different. This is an equivocation.

Now I'll weigh whether this rises to refutation. The claim as literally stated ("SSL combined with KD is a proven, top-tier approach for action spotting; COMEDIAN achieved 73.10% tight mAP, validating teacher?student distillation as a strong direction") is: factually correct on the number, correct that COMEDIAN combines SSL+KD, but it (a) conflates two distinct meanings of "distillation," making "validating teacher?student distillation" an unsupported/misleading inference, and (b) overstates "top-tier" given it has been surpassed. The quote ("First AS-specific framework that combines SSL with KD") supports the existence of the method but does NOT support the conclusion that this validates the VLM-teacher distillation direction for the user's pipeline.

As the adversarial 3rd voter, the equivocation on "distillation" plus the dated "top-tier" claim is enough to refute on the standard of "unsupported by quote / overstated."

### 15. user (2026-06-06T17:37:16.303Z)

Structured output provided successfully

### 16. assistant (2026-06-06T17:37:24.388Z)

Verification complete. Returned `refuted=true` with high confidence via StructuredOutput.

Summary of the adversarial review: The claim's headline number (COMEDIAN, 73.10 tight mAP on SoccerNet) is factually accurate and the SSL+KD characterization matches the source. However, the claim is refuted on two grounds: (1) it equivocates on "distillation" ? COMEDIAN's KD is feature-bank self-distillation for transformer pretraining/initialization, not the cloud-VLM-teacher ? small-local-student task-label distillation the research question is actually evaluating, so "validating teacher->student distillation as a strong direction" is an unsupported inferential leap; and (2) "top-tier" is dated ? COMEDIAN was WACV-2024 SOTA but was surpassed by T-DEED (77.25 tight avg-mAP) in the SoccerNet 2024 challenge.